Diffusion Models." These models use machine learning algorithms to remove noise, such as Gaussian noise, from an input signal, such as an image, through a series of iterative steps until a clean sample resembling the original signal is achieved.

While these models are proficient at generating high-quality images, the slow reverse denoising process and large pixel space can be major challenges. The sequential nature of the process makes reverse denoising slow, and generating high-resolution images consumes significant memory. Therefore, training and using these models for inference can be challenging.

2. Dall-E 2

OpenAI has developed DALL-E (stylized as DALL·E) and DALL-E 2, which are deep learning models that can produce digital images from natural language descriptions known as "prompts." OpenAI introduced DALL-E in a blog post in January 2021, which utilizes a modified version of GPT-3 to generate images. In April 2022, OpenAI announced DALL-E 2, a successor designed to produce more realistic images at higher resolutions while also combining concepts, attributes, and styles.

DALL-E, an advanced deep learning model created by OpenAI, has the capability to generate diverse styles of digital imagery, including photorealistic imagery, paintings, and even emoji. The model can manipulate and rearrange objects within its images and can create novel compositions without explicit instructions. Thom Dunn from BoingBoing notes that DALL-E can accurately depict objects in plausible locations, even when asked to draw something as unconventional as a daikon radish sipping a latte or riding a unicycle. Additionally, the model can infer appropriate details and fill in the blanks without specific prompts, such as adding Christmas imagery to prompts related to the celebration.

DALL-E also exhibits a broad understanding of visual and design trends. It can produce high-quality images from various viewpoints with only rare failures. According to Mark Riedl, an associate professor at the Georgia Tech School of Interactive Computing, DALL-E can blend concepts to create new, creative outputs. Furthermore, the model's visual reasoning ability is sufficient to solve Raven's Matrices, which are commonly used to measure human intelligence.

DALL-E has emerged as one of the most popular tools for AI Art, along with others like Stable Diffusion and Midjourney. However, unlike Stable Diffusion, it does not require a heavy set up.